RIG-SMOOTH IMPLIES RIG-FLAT WITH REGULAR SEQUENCE PRESENTATIONS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 4470-4558, following the statement at 4459-4468.

Editorial note. The parenthetical (details omitted) at line 4505 is the standard identity identifying restricted power series over the completed localization with the completed localization of restricted power series; recorded in the card's obligations.

Editorial note. Steps the author leaves to the reader are flagged here for the reviewer: source line(s) 4505 carry an "omitted"/"follows easily"/"immediate from the definitions" phrase. Those sentences are retained verbatim; the displayed argument around them stays inside the counted span.

Allowed standard prerequisites (editorial).

- Rig-smooth base change and locality. lemma-base-change-rig-smooth-continuous and lemma-rig-smooth-axioms, cited at proof lines 4473-4477. Supplies that the localizations phi_{f} stay rig-smooth, reducing rig-flatness to the naive condition.
- Regular sequences on open subsets of V(J). algebra-lemma-regular-sequence-in-neighbourhood, cited at proof lines 4486-4489. Supplies that it suffices to check J_q generated by a regular sequence at primes maximal for not containing an ideal of definition.
- Rig-closed points after localization and descent. lemma-rig-closed-point-after-localization (4498-4502), algebra-lemma-flat-increases-depth and more-algebra-lemma-flat-descent-regular-ideal (4527-4528). Supplies the rig-closed prime p' and the faithfully flat descent of regular-sequence generation and flatness along the displayed diagram.
- Fibre regularity and Grothendieck regular sequence criterion. lemma-fibre-regular (4545-4548), more-algebra-lemma-quotient-sequence-regular (4550-4552), algebra-lemma-grothendieck-regular-sequence (4553-4555). Supplies that f_1,...,f_m map to a regular sequence in the fibre, hence in the local ring, with flat quotient.

Source reference. Original reference 00LM: focused-seeds/geometry/S9-restricted-rig-smooth-rig-flat/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 00MG: focused-seeds/geometry/S9-restricted-rig-smooth-rig-flat/ref-tags.txt, source lines 6–6. Navigation pointer only: the

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Source reference. Original reference 061L: focused-seeds/geometry/S9-restricted-rig-smooth-rig-flat/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GCL: sources/stacks-restricted/pinned-original-source.tex, source lines 4371–4371. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GAJ: sources/stacks-restricted/pinned-original-source.tex, source lines 843–843. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GGJ: sources/stacks-restricted/pinned-original-source.tex, source lines 3604–3604. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GGG: sources/stacks-restricted/pinned-original-source.tex, source lines 3488–3488. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0GCK: sources/stacks-restricted/pinned-original-source.tex, source lines 4253–4253. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 068N: focused-seeds/geometry/S9-restricted-rig-smooth-rig-flat/ref-tags.txt, source lines 8–8. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 0GEE: focused-seeds/geometry/S9-restricted-rig-smooth-rig-flat/ref-tags.txt, source lines 9–9. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $\varphi : A \to B$ be an arrow of WAdm^{Noeth}. If φ is rig-smooth, then φ is rig-flat, and for any presentation $B = A\{x_1, \ldots, x_n\}/J$ and prime $J \subset \mathfrak{q} \subset$*

$A\{x_1, \ldots, x_n\}$ not containing an ideal of definition the ideal $J_{\mathfrak{q}} \subset A\{x_1, \ldots, x_n\}_{\mathfrak{q}}$ is generated by a regular sequence.

Proof. Let $f \in A$. To prove that φ is rig-flat we have to show that $\varphi_{\{f\}} : A_{\{f\}} \to B_{\{f\}}$ is naively rig-flat. Now either by viewing $\varphi_{\{f\}}$ as a base change of φ and using Lemma 0GCL or by using the fact that being rig-smooth is a local property (Lemma 0GCK) we see that $\varphi_{\{f\}}$ is rig-smooth. Hence it suffices to show that φ is naively rig-flat.

Choose a presentation $B = A\{x_1, \ldots, x_n\}/J$. In order to check the second part of the lemma it suffices to check $J_{\mathfrak{q}} \subset A\{x_1, \ldots, x_n\}_{\mathfrak{q}}$ is generated by a regular sequence for $J \subset \mathfrak{q}$ for $\mathfrak{q}$ maximal with respect to not containing an ideal of definition, see Algebra, Lemma 061L (which shows that the set of primes in $V(J)$ where there is a regular sequence generating J is open). In other words, we may assume $\mathfrak{q}$ is rig-closed in $A\{x_1, \ldots, x_n\}$. And to check that B is naively rig-flat, it also suffices to check that the corresponding localizations $B_{\mathfrak{q}}$ are flat over A.

Let $\mathfrak{q} \subset A\{x_1, \ldots, x_n\}$ be rig-closed with $J \subset \mathfrak{q}$. By Lemma 0GGG we may choose an $f \in A$ mapping to a unit in $A\{x_1, \ldots, x_n\}/\mathfrak{q}$ and such that the prime ideal $\mathfrak{p}'$ in $A_{\{f\}}$ induced is rig-closed. Below we will use that $A_{\{f\}}\{x_1, \ldots, x_n\} = A\{x_1, \ldots, x_n\}_{\{f\}}$; details omitted. Consider the diagram

$$\begin{array}{ccc} A\{x_1, \ldots, x_n\}_{\mathfrak{q}}/J_{\mathfrak{q}} & \longrightarrow & A_{\{f\}}\{x_1, \ldots, x_n\}_{\mathfrak{q}'}/JA_{\{f\}}\{x_1, \ldots, x_n\}_{\mathfrak{q}'} \\ \uparrow & & \uparrow \\ A\{x_1, \ldots, x_n\}_{\mathfrak{q}} & \longrightarrow & A_{\{f\}}\{x_1, \ldots, x_n\}_{\mathfrak{q}'} \\ \uparrow & & \uparrow \\ A & \longrightarrow & A_{\{f\}} \end{array}$$

The middle horizontal arrow is faithfully flat as it is a local homomorphism of local rings and flat as $A_{\{f\}}\{x_1, \ldots, x_n\}$ is the completion of a localization of the Noetherian ring $A\{x_1, \ldots, x_n\}$. Similarly the bottom horizontal arrow is flat. Hence to show that $J_{\mathfrak{q}}$ is generated by a regular sequence and that $A \to A\{x_1, \ldots, x_n\}_{\mathfrak{q}}/J_{\mathfrak{q}}$ is flat, it suffices to prove the same things for $JA_{\{f\}}\{x_1, \ldots, x_n\}_{\mathfrak{q}'}$ and $A_{\{f\}} \to A_{\{f\}}\{x_1, \ldots, x_n\}_{\mathfrak{q}'}/JA_{\{f\}}\{x_1, \ldots, x_n\}_{\mathfrak{q}'}$. See Algebra, Lemma 00LM or More on Algebra, Lemma 068N for the statement on regular sequences. Finally, we have already seen that $A_{\{f\}} \to B_{\{f\}}$ is rig-smooth. This reduces us to the case discussed in the next paragraph.

Let $\mathfrak{q} \subset A\{x_1, \ldots, x_n\}$ be rig-closed with $J \subset \mathfrak{q}$ such that moreover $\mathfrak{p} = A \cap \mathfrak{q}$ is rig-closed as well. By the characterization of rig-smooth algebras given in Lemma 0GAJ after reordering the variables $x_1, \ldots, x_n$ we can find $m \geq 0$ and $f_1, \ldots, f_m \in J$ such that

(1) $J_{\mathfrak{q}}$ is generated by $f_1, \ldots, f_m$, and
(2) $\det_{1 \leq i,j \leq m}(\partial f_j/\partial x_i)$ maps to a unit in $A\{x_1, \ldots, x_n\}_{\mathfrak{q}}$.

By Lemma 0GGJ the fibre ring

$$F = A\{x_1, \ldots, x_n\} \otimes_A \kappa(\mathfrak{p})$$

is regular. Observe that the A-derivations $\partial/\partial x_i$ extend (uniquely) to derivations $D_i : F \to F$. By More on Algebra, Lemma 0GEE we see that $f_1, \ldots, f_m$ map to

a regular sequence in $F_{\mathfrak{q}}$. By flatness of $A \to A\{x_1, \ldots, x_n\}$ and Algebra, Lemma 00MG this shows that $f_1, \ldots, f_m$ map to a regular sequence in $A\{x_1, \ldots, x_m\}_{\mathfrak{q}}$ and the quotient by these elements is flat over A. This finishes the proof. $\square$